10-4. CATE: The R-Learner

A consolidated, mathematical lecture companion

English notes based on the lecture slides *CATE: The R-Learner* (Prof. Sanghack Lee, GSDS SNU), following Nie and Wager (2021, Biometrika)

The central question is how to estimate the conditional average treatment effect with a loss function whose minimiser is the CATE itself, so that errors in the nuisance regressions affect the estimate only at second order.

These notes consolidate the slides, derive Robinson's residualised representation and the residual-on-residual interpretation, state the cross-fitted algorithm, prove Neyman orthogonality of the R-moment, explain the quasi-oracle property, and compare with the DR-learner. Physical PDF page numbers are listed in Section 6.

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1 Motivation and identification

Purpose. Set up the target and the residualised representation.

We observe i.i.d. $O_i = (X_i, Z_i, Y_i)$ with $X_i \in \{0, 1\}$ and want $\tau(z) = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = z]$. S-, T- and X-learners are easy to explain but can pass nuisance-regression bias into $\hat{\tau}$. DR-learners build a pseudo-outcome from a Neyman-orthogonal DR score and regress it on Z. The R-learner instead builds a *loss function whose target is $\tau(\cdot)$ itself* (Nie and Wager 2021).

Assumptions: consistency $Y_i = Y_i(X_i)$, unconfoundedness $\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp X_i \mid Z_i$, overlap $0 < e(Z_i) < 1$. **Nuisances:**

$$e(z) = \Pr(X = 1 \mid Z = z), \qquad m(z) = \mathbb{E}[Y \mid Z = z],$$

where m is the Z-only outcome mean, not an arm-specific mean μ_x. Residuals: $\tilde{Y}_i = Y_i - m(Z_i)$, $\tilde{X}_i = X_i - e(Z_i)$. The R-learner uses

$$\tilde{Y}_i = \tilde{X}_i \, \tau(Z_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0,$$

Robinson's partially linear decomposition with a varying coefficient.

Derivation. Fix $Z = z$ and let $\mu_x(z) = \mathbb{E}[Y(x) \mid Z = z]$, $\tau(z) = \mu_1(z) - \mu_0(z)$. By consistency and unconfoundedness, $\mathbb{E}[Y \mid X = x, Z = z] = \mu_0(z) + x\,\tau(z)$. Since m averages over the observed treatment mix,

$$m(z) = \mathbb{E}[Y \mid Z = z] = e(z)\mu_1(z) + \{1 - e(z)\}\mu_0(z) = \mu_0(z) + e(z)\tau(z),$$

so $\mathbb{E}[Y - m(Z) \mid X, Z = z] = \{X - e(z)\}\tau(z)$, i.e. $\mathbb{E}[\tilde{Y} \mid X, Z = z] = \tilde{X}\tau(z)$; the remainder is ε.

2 The R-loss

Purpose. Turn the representation into a loss and interpret its minimiser.

Oracle R-loss. Since $\mathbb{E}[\tilde{Y} \mid X, Z] = \tilde{X}\tau(Z)$, an oracle knowing m and e could estimate τ by minimising

$$L_R(\tau) = \mathbb{E}\big[\{\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\}^2\big], \qquad \tilde{\tau} = \arg\min_{\tau} \Big\{ \frac{1}{n} \sum_{i=1}^{n} \big(\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\} \Big\},$$

with a penalty/regulariser Λ_n (Robinson 1988; Nie and Wager 2021, §2).

A residual regression. Conditionally on $Z = z$ the oracle loss is $Q_z(t) = \mathbb{E}[(\tilde{Y} - \tilde{X}t)^2 \mid Z = z]$, a least-squares regression of $\tilde{Y}$ on $\tilde{X}$ through the origin. Its slope solves $0 = \mathbb{E}[\tilde{X}(\tilde{Y} - \tilde{X}t_z) \mid Z = z]$, so

$$t_z = \frac{\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z]}{\mathbb{E}[\tilde{X}^2 \mid Z = z]}.$$

Why this slope is the CATE. By iterated expectations, $\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z] = \mathbb{E}[\tilde{X}\,\mathbb{E}[\tilde{Y} \mid X, Z = z] \mid Z = z] = \mathbb{E}[\tilde{X}^2 \mid Z = z]\tau(z)$, hence $t_z = \tau(z)$. *The CATE is the population slope of outcome residuals on treatment residuals.*

3 Algorithm

Purpose. Implement the R-learner with cross-fitting.

1. Split the data into K folds (cross-fitting: each unit's residual uses nuisance models trained on other folds).
2. For each fold k, fit nuisances on the other folds: $\hat{m}^{(-k)}(z) \approx \mathbb{E}[Y \mid Z = z]$, $\hat{e}^{(-k)}(z) \approx \Pr(X = 1 \mid Z = z)$.
3. Out-of-fold residuals: $Y_i^r = Y_i - \hat{m}^{(-k_i)}(Z_i)$, $X_i^r = X_i - \hat{e}^{(-k_i)}(Z_i)$.
4. Estimate $\hat{\tau} = \arg\min_\tau \Big\{ \frac{1}{n} \sum_{i=1}^n \big(Y_i^r - X_i^r \tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\} \Big\}$.

Workflow: data $(X, Z, Y) \to$ cross-fit $\hat{m}, \hat{e} \to$ residualise $Y - \hat{m}(Z)$, $X - \hat{e}(Z) \to$ minimise the R-loss $\to \hat{\tau}(z)$. First-stage prediction creates residualised signals; the second stage learns τ.

The R-loss weights residual treatment variation.

$$\sum_i \big(Y_i^r - X_i^r \tau(Z_i)\big)^2 = \sum_i (X_i^r)^2 \Big(\frac{Y_i^r}{X_i^r} - \tau(Z_i)\Big)^2.$$

Units with large $|X_i^r|$ (treatment poorly predicted by Z_i) carry information about $\tau(Z_i)$; units with small $|X_i^r|$ (treatment nearly determined by Z_i) put little pressure on the local $\tau(Z_i)$. R-learners learn the CATE where residual treatment variation remains.

What can optimise the R-loss? The second stage is empirical risk minimisation, $\min_{\tau \in \mathcal{F}} \frac{1}{n} \sum_i (Y_i^r - X_i^r \tau(Z_i))^2 + \Lambda_n\{\tau\}$: simple choices (linear or sparse linear $\tau(z)$, splines, kernels) or flexible ones (boosting, random forests, neural networks, any learner with a squared-loss interface, using weights $(X_i^r)^2$ and targets Y_i^r / X_i^r). The learner for τ is tuned by *validation R-loss*, not by outcome prediction loss.

4 Orthogonality and the quasi-oracle property

Purpose. Show why nuisance errors matter only at second order.

A population moment is an expectation equation that is zero in the target population (before finite-sample fluctuation). For a candidate τ, use

$$\psi(O; \tau, m, e) = \{Y - m(Z) - (X - e(Z))\tau(Z)\}\{X - e(Z)\}, \qquad \mathbb{E}[\psi(O; \tau^*, m, e) \mid Z = z] = 0 \text{ at the truth.}$$

Proposition 4.1 (Neyman orthogonality)**.** *Let $\hat{m} = m + \delta_m$, $\hat{e} = e + \delta_e$ and hold $\tau = \tau^*$. Then*

$$\mathbb{E}[\psi(O; \tau^*, \hat{m}, \hat{e}) \mid Z = z] = \underbrace{\delta_m(z)\delta_e(z)}_{\textit{product}} - \underbrace{\tau^*(z)\delta_e(z)^2}_{\textit{square}}.$$

Proof. With $\tilde{Y} = Y - m(Z)$, $\tilde{X} = X - e(Z)$, the moment equals $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^* - \delta_m + \delta_e\tau^*)(\tilde{X} - \delta_e) \mid Z]$. The term $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^*)\tilde{X} \mid Z] = \mathbb{E}[\varepsilon\tilde{X} \mid Z] = 0$; the term $-\delta_e\mathbb{E}[\tilde{Y} - \tilde{X}\tau^* \mid Z] = -\delta_e\mathbb{E}[\varepsilon \mid Z] = 0$; and $(-\delta_m + \delta_e\tau^*)\mathbb{E}[\tilde{X} - \delta_e \mid Z] = (-\delta_m + \delta_e\tau^*)(-\delta_e)$ since $\mathbb{E}[\tilde{X} \mid Z] = 0$. □

No first-order δ_m or δ_e term remains: small nuisance errors create only second-order bias, so flexible nuisance fits are usable (Nie and Wager 2021; Chernozhukov et al. 2018).

The quasi-oracle property. Measure excess risk with the oracle population R-risk $R(\tau) = \mathbb{E}[\{Y - m(Z) - (X - e(Z))\tau(Z)\}^2]$, via $R(\tau) - R(\tau^*)$. Let $\hat{\tau}$ fit residuals from $\hat{m}, \hat{e}$ and $\tilde{\tau}$ the oracle version with true m, e. Nie and Wager show

$$R(\hat{\tau}) - R(\tau^*) \approx R(\tilde{\tau}) - R(\tau^*),$$

provided the nuisances are estimated well enough (RMSE $o(n^{-1/4})$, so their product is $o(n^{-1/2})$). The rate depends mainly on the complexity of τ: after residualisation, the second stage learns τ, not the full complexity of m and e.

Why cross-fitting matters. If observation i helps train $\hat{m}, \hat{e}$, then $\hat{m}(Z_i), \hat{e}(Z_i)$ can memorise noise in (Y_i, X_i): residuals become too small, the noise fitted by the nuisances leaks into the R-loss, and the second stage may mistake first-stage overfitting artefacts for heterogeneity in $\tau(Z)$. With cross-fitting, residuals are out-of-fold predictions, unit i's noise was not used to train its nuisances, and orthogonality can remove first-order nuisance bias. Orthogonality handles nuisance error; cross-fitting prevents reusing the same noise twice.

5 Comparison and summary

Purpose. Contrast the R-learner with the DR-learner.

	DR-learner	R-learner
Second stage	regress $\hat{\phi}_i^{DR}$ on Z_i	minimise a residualised weighted loss
Orthogonality	via the EIF / DR score	via Robinson residualisation
Main nuisances	μ_0, μ_1, e	m, e
Model selection	validation MSE for $\hat{\phi}^{DR}$	validation R-loss

DR pseudo-outcome:

$$\hat{\phi}_i^{DR} = \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i) + \frac{X_i\{Y_i - \hat{\mu}_1(Z_i)\}}{\hat{e}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{\mu}_0(Z_i)\}}{1 - \hat{e}(Z_i)}.$$

Validation losses on a held-out set V: DR, $\sum_{i \in V}\{\hat{\phi}_i^{DR} - \hat{\tau}(Z_i)\}^2$; R, $\sum_{i \in V}\{Y_i^r - X_i^r \hat{\tau}(Z_i)\}^2$.

What the R-learner is for. Residualised CATE estimation (estimate $\tau(Z)$ after removing prognosis $m(z)$ and assignment $e(z)$); model selection (tune and compare candidate τ models by validation R-loss); regularised heterogeneity (put shrinkage or structure directly on τ when the effect signal is weak).

Summary.

- R-learning starts from Robinson's decomposition $Y - m(Z) = \{X - e(Z)\}\tau(Z) + \varepsilon$.
- Estimate m and e by cross-fitting, then minimise the residualised R-loss.
- The R-loss is orthogonal to nuisance error and can be optimised with many ML tools.
- The quasi-oracle result says the difficulty lies mainly in the complexity of τ, not of m and e.

6 Source coverage

Topic	Physical pages
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